

times a few advanced capabilities too), a good battery back-up and is easy to use.

In the labs. People in the labs use not only benchtops but also handheld instruments due to a variety of reasons. One is the commercial aspect. The price of the handheld is usually much lower than its benchtop counterpart. Next is the space constraint. If you stack benchtops, these will take up huge rack space, whereas the same functionalities integrated into a single portable instrument occupies much lesser space. Some of the lab-testing scenarios may require the portability feature, and handhelds would be useful there. But you also have to look at the performance of the device when it comes to lab tests, whether parameters of the device matches your test requirements. There are cases in lab where you do not require performing in-depth testing—this is where portable instruments come in.

What to consider

Here are a few attributes to look for when purchasing a new handheld device for your application.

Form factor and portability. This is the most important factor to be considered while selecting a handheld device. If the instrument is to be carried around all the time, customers prefer maximum number of functions integrated into a single system with smallest form factor. Some examples are Tektronix RSA306 spectrum analyser that weighs about 0.59kg and has a dimension of just 30.5x190.5x127mm, and R&S FSH series spectrum analysers that have a size of only about 194 × 300 × 144mm and weigh 2.5kg.

Simple UI. People who use handhelds in fields are usually not research and development engineers. Mostly field engineers and technicians use such devices and their expertise is different. They prefer instruments that perform test and measurement in very few set-ups. According to a survey by Frost & Sul-



MT8220T BTS Master, a multi-function handheld durable tester

livan, “Engineers are increasingly associating the concept of a UI and user experience with the one they use on their consumer electronics devices.” This means that users expect touch based UIs that respond to gestures like pinch-zoom, tap, etc. So, it is best to buy a device that has a very simple and intuitive UI.

Application-specific integration. One of the major advantages of handheld instruments is that these combine more than one test-and-measurement instruments into a single portable device. But the challenge is to make a simple UI that has the controls and displays for all instruments in one place.

Many of the handheld devices introduced in the market these days are application-specific. Almost all instruments required for one type of application, say, cable testing, are integrated into a single box. Network Master Flex/Pro network field tester and BTS Master for base-station testing by Anritsu, and FieldFox RF and microwave analyser series from Keysight are examples. You can also find some of the accessories, such as a calibration kit, built into FieldFox, thus reducing the number of instruments to be carried around.

“Even if the device is not application-specific, it is possible for the modern graphic capabilities of frameworks like Windows Presentation Foundation (WPF) to build visually-stunning and extremely-intuitive GUI that combines all instruments into a single window,” says Chinmay Misra of NI.

Modes of operation. In case you have to work on multiple projects, go for a platform that supports other applications as well. Suppose the application the user is currently working on requires a cable tester and spectrum analyser. But after a few months, they get a new application that may require a power meter and data-network analyser. So, instead of buying a separate handheld for the new application, the user can get a software upgrade to the existing instrument, which lets him or her use the power meter and data-network analyser features. There is no calibration or hardware change required as a fully-capable hardware is directly given to the user initially.

Battery life. All portable devices need not be battery-operated. Deeper integration of multiple capabilities and high-speed digital technologies in a single device calls for outstanding